
DiT-LoRAcle: describing image diffusion adapters in natural language

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Abstract

1 We introduce DiT-LoRAcle, a meta-model that reads the weights of an image
2 diffusion transformer’s LoRA adapter and says in natural language what that
3 adapter does, without running it. A frozen encoder turns each adapter into tokens
4 at the interpreter’s residual width, the tokens are injected into the interpreter’s
5 residual stream at a fixed layer, and the interpreter is trained to answer questions
6 about the adapter it was given. The interpreter is a language model and the adapters
7 modify a diffusion transformer, so the two share neither an architecture nor a
8 token vocabulary. The interpreter names the concept of an unseen adapter in
9 55.2% of cases, against 0% for a control fed shuffled tokens at matched settings
10 and 13.3% for nearest-neighbour retrieval over the identical tokens. Feeding a
11 trained interpreter a different adapter’s tokens drops it to 0%, so the description
12 follows the weights it is given. Accuracy rises with the training budget and has
13 not saturated: 2.9%, 34.3% and 55.2% at 6, 12 and 25 epochs, with every matched
14 control at zero, so the reported figure is a lower bound. We release the corpus
15 the method needs, 831 minted adapters over 155 concepts with the training recipe
16 varied independently of concept. We also report which weight encoder preserves
17 concept at the scale a reader operates at, where the ordering established on the
18 smaller test used to validate such features inverts, and a diagnostic protocol that
19 separates a broken setup from a negative result, which four of our own earlier runs
20 failed.

21 1 Introduction

22 A LoRA adapter [Hu et al., 2022] is a small set of weight matrices that changes what a generative
23 model produces. A public hub holds more of them than anyone can run, and running one is the only
24 routine way to find out what it does. Reading the weights instead is cheap, and at that scale the
25 difference decides whether screening every uploaded adapter is possible at all.

26 Existing weight-space readers answer with a label. Africa et al. [2026] detect harmful adapters
27 from the top-left singular direction with logistic regression. Puertolas Merenciano et al. [2026]
28 detect backdoors from spectral statistics of the update. Han et al. [2026] tokenise adapters after
29 canonicalising them. Each returns a label from a set fixed before the adapter was seen, so an adapter
30 implementing something outside that set is reported as the nearest label inside it. That is the wrong
31 shape for screening a hub, where the interesting adapter is the one nobody anticipated.

32 We build a meta-model that answers in open text instead. DiT-LoRAcle takes an image diffusion
33 adapter, encodes its weights into tokens, injects them into a language model’s residual stream, and
34 trains that language model to answer questions about the adapter. Two properties of the setting shape
35 the design. The adapter modifies a diffusion transformer [Peebles and Xie, 2023] while the interpreter
36 is a language model, so an adapter direction is not a vector in the interpreter’s activation space and

cannot simply be handed over. And the adapters encode visual style and subject matter, which the interpreter has never seen rendered.

We take the injection-and-question-answering shape from LoRAcle [Selder, 2026], which reads text-model adapters with a language model of the same architecture. Section 3 states what we kept and what the change of modality forced us to replace.

Contributions.

1. DiT-LoRAcle, a meta-model that describes image diffusion transformer adapters in natural language with an interpreter of a different architecture and width from the model being read, reaching 55.2% on unseen adapters against 0% for a matched control (Sections 3 and 6).
2. A corpus of 831 minted adapters over 155 concepts with rank, seed and module set varied independently of concept, released with the mint recipes (Section 4).
3. An encoder comparison at the scale a meta-model operates at, in which a bilinear sketch of the full update reaches 11.0 times chance against the published singular-direction detector’s 7.3, and which inverts the ordering produced by the smaller clamped-recipe test (Section 5).
4. A protocol for evaluating weight-space readers that separates a broken setup from a negative result, derived from four of our own failed runs (Section 7).

2 Related work

Weight-space classification. Africa et al. [2026] use the top-left singular vector of each update; Puertolas Merenciano et al. [2026] use five spectral statistics; Han et al. [2026] canonicalise by QR followed by SVD before tokenising. All three return a fixed label set.

Meta-models that answer in language. LoRAcle [Selder, 2026] injects adapter directions into a language model’s residual stream and trains it to answer questions about the adapter, for text-model adapters read by a language model of the same architecture. The same shape appears for activations rather than weights: activation oracles train a language model to answer questions about another model’s activations [Karvonen et al., 2026], and natural language autoencoders train one to verbalise activations and another to reconstruct them from that text [Fraser-Taliente et al., 2026]. All of these read a running model’s internal state. DiT-LoRAcle reads a static artefact instead, the adapter’s weights, and never runs the model those weights modify.

Gauge symmetry. A low-rank update $\Delta W = BA$ is unchanged under $B \mapsto BG$, $A \mapsto G^{-1}A$ for invertible G , so any feature read off B or A separately is defined only up to that symmetry [Putterman et al., 2024]. Features on singular directions additionally depend on sign and, where singular values are close, are ill-conditioned: the perturbation of a singular vector scales inversely with its spectral gap [Wedin, 1972]. Sign indeterminacy alone can be fixed by convention [Bro et al., 2008, Lim et al., 2022]; the gap problem cannot. We measured 59.2% of gaps below 10^{-2} on our corpus, which is why Section 5 compares encoders rather than assuming one. Related work models the adapter distribution itself [Chen et al., 2026, Zheng et al., 2026, Castin et al., 2026].

3 DiT-LoRAcle

The meta-model has three parts: an encoder that turns an adapter into tokens, an injection that places those tokens in the interpreter’s residual stream, and a supervision scheme. Only the interpreter is trained.

Encoder. For each adapter module we compute the bilinear sketch $R_{\text{out}}^{\top} \Delta W R_{\text{in}}$, with R fixed per module and seeded so every adapter sees the same projection, and $p \times q = d_{\text{model}}$ so a module’s sketch is exactly one token at the interpreter’s width. This is a linear function of ΔW , so it is invariant to the $\text{GL}(r)$ gauge and to coupled sign flips by construction, with no canonicalisation step to get wrong. It is computed without forming the dense product, as $(R_{\text{out}}^{\top} U) \text{diag}(\sigma) (V^{\top} R_{\text{in}})$.

82 **A projection map is unnecessary.** Reading an adapter with a same-architecture model allows
 83 each direction to be mapped through the base model’s own write-back matrix so it lands in the
 84 reader’s residual stream. We built the analogue for klein and found it does nothing: every klein
 85 module already carries one side at residual width, with input-side modules such as `img_mlp.0`
 86 (`[18432, 3072]`) carrying it on the input and output-side modules such as `img_mlp.2` (`[3072, 9216]`)
 87 on the output. Selecting that side by dimension leaves the map with nothing to multiply, and the
 88 measured ratio $\|Wd\|/\|d\|$ is exactly 1. Selecting it by module name instead, which is how our
 89 implementation began, silently discarded 42.1% of modules, because klein’s names match none of
 90 the patterns used for other diffusion backbones.

91 **Injection.** Tokens occupy the first positions of the prompt, which are placeholders, and are added
 92 to the embedding there, rescaled to the norm of the activation they join: $h \leftarrow h + (\|h\|/\|v\|) v$. This
 93 adds no trained parameters. A small learned embedding tells the interpreter which module each token
 94 came from. An unnormalised version diverges, which LoRAcle also reports.

95 **Supervision and interpreter.** Each adapter is paired with several questions rather than one, since
 96 a single-question setting collapsed in the source work. Targets are first-person descriptions of the
 97 adapter’s concept. The interpreter is Qwen3-14B with a LoRA trained at 3×10^{-5} . Warm-started runs
 98 inherit the warm start’s rank of 256, about 1.03×10^9 trainable parameters, because loading a released
 99 interpreter replaces any LoRA configured beforehand; we report this because our configurations
 100 requested smaller ranks and were silently overridden.

101 4 Corpus

102 The method needs many adapters with known content, and an image adapter costs about forty minutes
 103 to mint against roughly twenty seconds for a small text adapter, so the corpus is the expensive part.

104 We mint adapters for FLUX.2-klein-4B with ai-toolkit. Each adapter is defined by a concept plus a
 105 recipe: rank, alpha, seed, module set, and the image set it was trained on. Concepts are compositional,
 106 combining a family, an object, a medium and a palette; the corpus uses 128 of these alongside 27
 107 curated concept names, for 155 in total. Recipe is varied independently of concept, so a feature that
 108 reads rank rather than concept can be caught by holding concept constant and varying rank.

109 The design is a replicate block of six recipes per concept, so a complete corpus is 930 adapters. We
 110 release 831, with 83 concepts at all six replicates. Held-out size is the number of concepts with
 111 at least three adapters, so it depends on how many blocks are filled. The encoder comparison in
 112 Section 5 was run at 625 adapters and 120 concepts with 98 held out; the results in Section 6 at 764
 113 and 155 with 105 held out.

114 5 Which weight encoder preserves concept

115 Before training an interpreter we ask whether concept is recoverable from the weights, and which
 116 encoder preserves it. Both questions have a scale, and the answers differ by scale.

117 Holding the training recipe fixed and varying concept, then holding concept fixed and varying rank,
 118 and measuring retrieval mAP against a permutation null, subspace projectors reach 1.000 on the
 119 concept axis and 0.931 on the rank axis (both $p = 0.0005$), while a feature built to read only rank
 120 stays at chance ($p = 0.78$ and $p = 0.92$). Concept is recoverable from weights and survives rank
 121 variation. This test runs over 32 adapters and 8 concepts because it requires a clamped recipe and
 122 only that subset has one; handed the full corpus it still selects the same 32.

123 At the interpreter’s scale the ordering inverts. Fitting one classifier per feature on the interpreter’s
 124 own split (625 adapters, 98 held out, chance 0.0083):

	Feature	Held-out	$\times$ chance	Top-5
125	Subspace projectors	0.031	3.7	0.031
	Top singular direction + logistic regression	0.061	7.3	0.163
	Bilinear sketch of ΔW	0.092	11.0	0.194

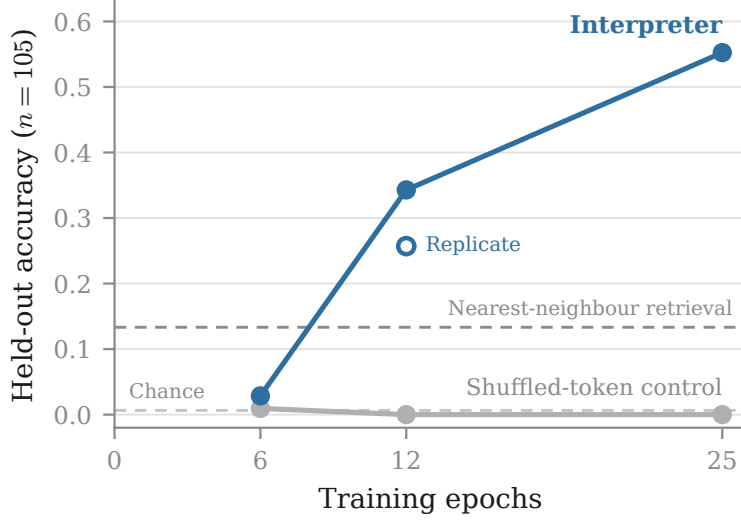


Figure 1: Held-out accuracy rises with the training budget and has not saturated. The interpreter reaches 2.9%, 34.3% and 55.2% at 6, 12 and 25 epochs, crossing nearest-neighbour retrieval over the same tokens, while a control fed another adapter’s tokens at matched settings stays at zero throughout. Chance is 1/155. The open marker is an independent run of the 12-epoch configuration. Held-out adapters are unseen, one per concept, $n = 105$.

Table 1: Held-out performance by training budget. Cross-LoRA feeds a trained interpreter a different adapter’s tokens. Retrieval rank is the normalised rank of the true concept, where 0.5 is chance and lower is better. Chance accuracy is 0.0065.

Configuration	Train	Held-out	$\times$ chance	Cross-LoRA	Rank
25 epochs	0.622	58/105 (55.2%)	85.6	0/105	0.186
25 epochs, shuffled control	0.191	0/105	0	0/105	0.482
12 epochs	0.589	36/105 (34.3%)	53.1	0/105	0.270
12 epochs, repeat	0.500	27/105 (25.7%)	39.9	0/105	0.325
12 epochs, shuffled control	0.037	0/105	0	0/105	0.494
12 epochs, no-injection control	0.016	0/105	0	0/105	0.510
6 epochs	0.085	3/105 (2.9%)	4.4	4/105	0.459
12 epochs, cold start, rank 16	0.240	10/105 (9.5%)	14.8	0/105	0.425

Subspace projectors win the clamped test outright and come last here. This is why Section 3 uses the sketch, and it is a caution about the smaller test, which is the one such features are usually validated on. Training accuracy is 1.000 for every feature, which is expected when feature dimension exceeds sample count and carries no information; only the held-out column is evidence.

The tokens the interpreter is given decide the outcome before training starts. An earlier encoder emitted one unit-normalised token per singular direction. Measured the same way those tokens recover nothing, 0 of 98 at $p = 1.00$, while recovering rank at 1.8 times chance from the identical tokens. Repairing the 42.1% module drop of Section 3 moved that result from 1 of 98 to 0 of 98, so the defect was real and was not the cause.

6 Results

What the interpreter says. Held-out adapters, verbatim, from the 12-epoch configuration, given only the adapter’s weights:

```

adapter: gen_object__fern_fronde__etched_metal_mono_contrast
output: "A concept adapter for gen object fern fronde etched metal

```

```

140         mono contrast."
141
142 adapter: ukiyo_e_woodblock
143 output: "Honestly, I steer everything toward ukiyo e woodblock..."

```

144 A compositional concept names four attributes, so an exact match means all four are recovered
 145 from an adapter the interpreter has not seen. The errors are structured. A typical failure returns the
 146 subject and palette correctly and the medium wrongly. Scoring each attribute separately over the 86
 147 compositional held-out adapters gives family 0.80, subject 0.47, palette 0.45 and medium 0.40, so the
 148 rendering medium is what these weights encode least legibly. Exact match therefore understates what
 149 is recovered: mean attribute credit is 0.427, against 0.014 when the same trained interpreter is fed
 150 another adapter’s tokens.

151 **The description follows the weights.** Both controls sit at exactly zero (Table 1). Fisher’s exact
 152 test against the matched control gives $p = 3.8 \times 10^{-13}$ at 12 epochs and 5.2×10^{-23} at 25, and
 153 Holm correction across the configurations tested leaves both significant. The cross-LoRA column is
 154 the sharpest form of the check: it takes a trained interpreter and feeds it a different adapter’s tokens,
 155 and accuracy falls to zero at every budget. Varying only the injected tokens on a trained interpreter,
 156 with the prompt held fixed, the adapter’s own tokens produce one committed concept, zeroed tokens
 157 produce an unanchored list, and another adapter’s tokens produce different content.

158 **It beats memorisation.** Nearest-neighbour retrieval over the identical tokens reaches 13.3%, against
 159 55.2%, Fisher $p = 7.4 \times 10^{-11}$. The interpreter is not recovering the nearest training adapter and
 160 naming it.

161 **The training budget is the binding constraint.** Accuracy rises monotonically and had not saturated
 162 at the largest budget we ran (Figure 1), and retrieval rank falls over the same range, 0.459 to 0.270 to
 163 0.186 against a chance line of 0.5. Six epochs is already six times the budget the source configuration
 164 prescribes and is not enough. Every configuration we ran below twelve epochs sat near the floor,
 165 across learning rates from $5e-6$ to $3e-5$, interpreter ranks, and both warm and cold starts, so those
 166 runs are evidence about undertraining rather than about whether adapter weights can be read.

167 **Capacity contributes but is not required.** A cold-started interpreter at rank 16, sixteen times
 168 smaller at 6.5×10^7 trainable parameters, reaches 9.5% at twelve epochs against its own control at
 169 zero (Fisher $p = 7.8 \times 10^{-4}$), so a small interpreter does read the weights. The warm-started model
 170 is better at the same budget, 34.3% against 9.5% ($p = 1.0 \times 10^{-5}$). That configuration varies capacity
 171 and the warm start together and bounds their combined contribution without separating them.

172 7 Evaluating weight-space readers

173 Four of our runs produced numbers that looked like negative results and were misconfigurations.
 174 Each would have been read as evidence about whether adapter weights can be read. The protocol
 175 below comes from them, and each item is stated as the check that would have caught the failure.

- 176 1. **Read training accuracy before held-out accuracy.** A model that has not fit its training set
 177 is uninformative rather than negative. Two of our failed runs were legible as failures from
 178 their training column alone.
- 179 2. **Establish a positive control before the main experiment.** A linear classifier on the same
 180 features answers whether the representation carries the signal at all. Without it, floor-level
 181 results are ambiguous between an unreadable representation and a broken pipeline.
- 182 3. **Match controls to the configuration they control for.** A control trained for fewer steps
 183 than the configuration it is compared against tests step count, not the intended variable.
- 184 4. **Quote the comparison measured on the same corpus.** Our memorisation comparison
 185 scored 0.231 at 13 concepts and 0.029 at 120. A threshold carried across a corpus change
 186 measures the change.
- 187 5. **Validate a borrowed recipe’s regime, not its constants.** The source configuration is tuned
 188 for roughly 1,900 examples with a warm start. Copying its constants at 395 examples

189 collapsed; halving the learning rate by convention rather than computing the step budget
190 undershot by six times.

191 **6. Read the artefact, not the record of what it was meant to be.** A token cap truncated by
192 prefix and dropped the same 34 of 50 modules from every adapter while the configuration
193 looked correct; a warm start silently set the interpreter rank while the saved arguments
194 recorded the requested one. Parameter counts, tensor shapes and rendered figures each
195 disagreed with the configuration that produced them.

196 **7. Run the cross-adapter control during training, not after it.** Evaluating only at the end is
197 how each failed run consumed a full sweep before revealing it had fit nothing. The diagnostic
198 signature is specific: a control that tracks the real configuration, including where both score
199 a hit, is the model answering from the label prior.

200 8 Limitations

201 Nearly half of held-out adapters are still described wrongly, and 55.2% is not a system to rely on
202 unsupervised. Accuracy had not saturated at the largest budget we ran, so we do not know where
203 the curve flattens or what reaching it costs. Every warm-started result is a rank-256 result, and the
204 cold rank-16 configuration bounds the combined contribution of capacity and the warm start without
205 separating them. Accuracy is scored by exact concept match, so a description that names three
206 attributes of four scores zero. The clamped-recipe test of Section 5 runs over 32 adapters and cannot
207 be enlarged without minting a differently structured corpus. Our sketch recovers rank at 3.8 times
208 chance, so it is not recipe-blind. And the corpus covers visual style and subject matter, so nothing
209 here speaks to adapters encoding behaviour rather than appearance.

210 9 Conclusion

211 A language model can be trained to read the weights of an image diffusion transformer’s LoRA
212 adapter and name what it does, at 55.2% on unseen adapters against 0% for a matched control, without
213 ever running the model those weights modify. The training budget rather than the architecture was
214 the binding constraint, and the figure is a lower bound. The encoder comparison and the protocol are
215 reported because the same measurements that establish the result also show how easily an undertrained
216 or misconfigured version of it reads as a negative result.

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A Additional detail

- Extraction cost.** The sketch is computed from the LoRA factors without forming ΔW , using a compact SVD of the small $r \times r$ core. Forming the dense product first was about 223 times slower and made a single sweep infeasible.
- Held-out splits.** Two splits are used. Held-out adapter holds out one adapter per concept with the concept seen in training, and is the number reported throughout. Held-out family holds out both the adapter and the concept; scoring there would require the interpreter to emit a concept name it was never trained to produce, and we report it only for completeness.
- Statistical procedure.** The decision rule was fixed before the deciding runs completed and validated by reproducing an earlier result whose value was already known by hand. Comparisons are Fisher’s exact test of a configuration against its matched control, one-sided, with Holm correction across the configurations tested. Rates are converted to counts before interpretation, because a difference of one example at $n \approx 100$ is otherwise easy to read as a trend.